Information Sciences 277 (2014) 247–262

Contents lists available at ScienceDirect

Information Sciences

journal homepage: www.elsevier.com/locate/ins

Analysis and extension of multiresolution singular value decomposition

Gaurav Bhatnagar [a,*], Ashirbani Saha [b], Q.M. Jonathan Wu [b], Pradeep K. Atrey [c]

[a] Indian Institute of Technology Jodhpur, Rajasthan, India
[b] University of Windsor, ON, Canada
[c] University of Winnipeg, MB, Canada

ARTICLE INFO

Article history:
Received 24 October 2011
Received in revised form 27 October 2013
Accepted 9 February 2014
Available online 18 February 2014

Keywords:
Multiresolution analysis
Singular value decomposition
Wavelet transform
Randomness
Chaotic map

ABSTRACT

This paper presents an analysis of the multiresolution form of singular value decomposition SVD (MR-SVD) and generalizes the analysis into a random multiresolution form of singular value decomposition (R-MR-SVD) with an intrinsic randomness. The core idea behind the proposed transform is to introduce randomness in the computing process based on parameters without which one can neither decompose nor reconstruct the data correctly. The proposed transform inherits the excellent properties of MR-SVD along with its own unique features, which can be useful in many research areas. Image encryption, lossy image compression, and face recognition are the primary applications used to illustrate the practical usage of the proposed transform.

© 2014 Elsevier Inc. All rights reserved.

1. Introduction

In the past decade, numerous mathematical transforms have been introduced to analyze signals. According to the requirements of signal processing applications, researchers have periodically modified these transforms. For instance, the basic tool for signal processing and analysis is the Fourier Transform (FT), which tries to approximate the signal in terms of sinusoids. One of the major drawbacks of FT is that it is not applicable to non-stationary signals. To overcome this problem, researchers have proposed several alternatives, namely short-time Fourier Transform, Gabor Transform, wavelet transform, etc.

Among these alternatives, the most frequently used and the most powerful is the wavelet transform (DWT). Several modifications have been performed on wavelets to make them useful in different scenarios. The core idea behind all these methods is to combine the multiresolution representation with filters depending on a threshold parameter. However, a conceptually complex structure, a huge computational memory (due to the use of filters), and limiting-perfect reconstruction (the term limiting indicates that the reconstruction is accurate to a specified degree) somewhat limit the use of DWT. On reviewing the wavelet literature, one finds that perfect reconstruction can only be achieved when the wavelet is applied on an infinite sequence of data. Almost all existing wavelets have the property of perfect reconstruction for infinite data sequences but do not exhibit the same property for finite data sequences. As a possible solution to these issues, Sweldens [21] and Kakarala and Ogunbona [12] devised the lifting wavelet transform (LWT) and multiresolution singular value decomposition (MR-SVD), respectively. Generally, it is difficult to find the lifting scheme for every existing wavelet easily.

* Corresponding author. Tel.: +91 9610170658.
E-mail address: goravb.uwindsor@gmail.com (G. Bhatnagar).

http://dx.doi.org/10.1016/j.ins.2014.02.018
0020-0255/© 2014 Elsevier Inc. All rights reserved.

Conversely, the MR-SVD scheme is very simple and has perfect reconstruction. Hence, this work concentrates on exploring MR-SVD and generalizing it to a random MR-SVD (R-MR-SVD).

In the past few years, some applications have been identified for MR-SVD. These applications include speaker recognition [14], speech signal enhancement [22], image compression [2,3], watermarking [1,13], stereo vision [15], fault diagnosis [23], image fusion [16] and other miscellaneous applications [10,19]. Among the possible applications of the newly proposed transform (R-MR-SVD), the primary and perhaps most important application lies in the area of information security. In information security, randomness plays a remarkable role, as does R-MR-SVD, as it introduces randomness to the process. To illustrate the practical applicability of R-MR-SVD, the proposed transform is applied to image encryption, lossy compression, and face recognition. Experimental results indicate that R-MR-SVD performs better performance than MR-SVD and DWT, which further demonstrates the possibilities of R-MR-SVD in information security. In compression, R-MR-SVD has similar performance when compared to MR-SVD and DWT at the lower compression ratios, and the performance enhances as we increase the compression ratio. R-MR-SVD and MR-SVD have similar performance levels, but perform better than DWT in face recognition.

The remainder of the paper is organized as follows: the MR-SVD is reviewed in Section 2 and the proposed R-MR-SVD is introduced in Section 3. The properties and implementation details of the proposed transform are described in Section 4. Three possible applications to image encryption, lossy image compression, and face recognition are discussed in Sections 5–7 respectively. Finally, the concluding remarks are given in Section 8.

2. Multiresolution singular value decomposition (MR-SVD)

The singular value decomposition (SVD) [11] of a real (imaginary) valued $M \times N$ matrix A, where $M \leqslant N$, is the factorization $A = USV^T$ where U and V are *orthogonal (unitary) matrices* and S is a *diagonal matrix* given by $S = diag(\sigma_1, \sigma_2, \ldots, \sigma_r)$, where $\sigma_i, i = 1(1)r$ are the singular values of matrix A with $r \leqslant \min(M, N) = M$ and satisfying $\sigma_1 \geqslant \sigma_2 \geqslant \cdots \geqslant \sigma_r$. The first r columns of V are called *right singular vectors* and the first r columns of U are called *left singular vectors*. Motivated by the excellent properties of SVD, Kakarala generalized SVD to MR-SVD in 2001. The basic idea of their extension is to replace low and high pass filters (used in wavelet transform) with SVD. The detailed flow of MR-SVD is depicted in Fig. 1 and is described as follows:

Let $X = [x(i,j)]_{i\in[1,m],j\in[1,n]}$ represent a 2D finite signal of size $m \times n$ such that m and n are dyadic. The computation of MR-SVD corresponds to the following steps,

1. Segment X into non-overlapping blocks of size $p \times q$ and arrange each block into a $pq \times 1$ vector by stacking columns to form the matrix X_1 of size $pq \times mn/pq$.
2. Evaluate the corresponding centered matrix i.e. $\overline{X}_1$ followed by the scatter matrix (T_1) computation as $T_1 = \overline{X}_1\overline{X}_1^T$ of size $pq \times pq$.
3. Let U_1 be the modal matrix (matrix of eigenvectors) which diagonalize the scatter matrix T_1, i.e., $U_1^T T_1 U_1 = S_1^2$, where $S_1^2 = diag\left(s_1(1)^2, s_1(2)^2, \ldots, s_1(pq)^2\right)$ contains the squares of the singular values of T_1.
4. Construct matrix $\widehat{X}$ as $\widehat{X} = U_1^T\overline{X}_1$.

Approximation and detail parts can be obtained by stacking the rows of $\widehat{X}$ into an array. The first row of $\widehat{X}$ corresponds to the largest singular value and is hence considered the approximate part of the signal. Similarly, the remaining row of $\widehat{X}$ corresponds to the remaining singular values (in decreasing order) and is hence considered the detail parts of the signal i.e. $\phi_1 = \widehat{X}(1,:)$ and $\psi_1^i = \widehat{X}(i,:)$, where ψ_1^i is the ith detail sub-band at level 1.

3. Random multiresolution singular value decomposition (R-MR-SVD)

This section defines a random form of multiresolution SVD (R-MR-SVD). The R-MR-SVD can be derived from the essence of the MR-SVD. Since, R-MR-SVD is the generalization of the MR-SVD, it satisfies all the properties of MR-SVD. The core idea behind the generalization from MR-SVD to R-MR-SVD is to produce randomness in the process. Randomness is produced by

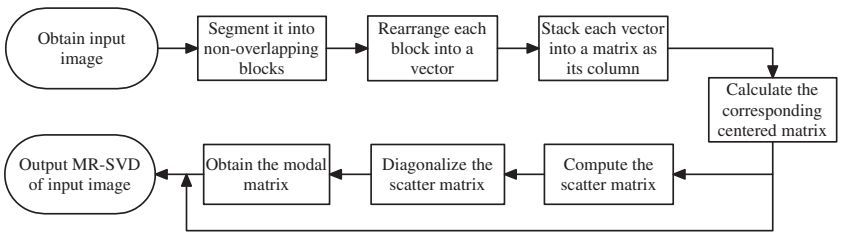

Fig. 1. Working flow of MR-SVD.

selecting two random matrices and is introduced in the fourth step of the MR-SVD. The computation of R-MR-SVD is as follows (the variables used are the same as in Section 2).

1. Segment X into non-overlapping blocks of size $p \times q$ and arrange each block into a $pq \times 1$ vector by stacking columns to form the matrix X_1 of size $pq \times mn/pq$.
2. Evaluate the corresponding centered matrix i.e. $\overline{X}_1$ followed by the scatter matrix (T_1) computation as $T_1 = \overline{X}_1\overline{X}_1^T$ of size $pq \times pq$.
3. Let U_1 be the modal matrix that diagonalizes the scatter matrix T_1, i.e., $U_1^T T_1 U_1 = S_1^2$, where $S_1^2 = diag\left(s_1(1)^2, s_1(2)^2, \ldots, s_1(pq)^2\right)$ contains the squares of the singular values of T_1.
4. Adopting keys K_1 and K_2 as the initial seeds to generate two invertible random matrices $\mathcal{K}_i = \{k|0 < k < 1, i = 1, 2\}$ of size $pq \times pq$ and $mn/pq \times mn/pq$ respectively.
5. Randomize matrix $\overline{X}_1$ as $\overline{X}_1^r = \mathcal{K}_1\overline{X}_1\mathcal{K}_2$.
6. Construct matrix $\widehat{X}$ as $\widehat{X} = U_1^T\overline{X}_1^r$.

Similar to MR-SVD, the approximation and detail parts for R-MR-SVD can be obtained by stacking rows of $\widehat{X}$ into an array where the first row corresponds to the approximate sub-band and remaining rows correspond to detail parts of the signal.

4. Analysis of MR-SVD and implementation of R-MR-SVD

In order to explore the performance of the proposed R-MR-SVD, the MATLAB platform is used and a number of experiments are performed on different 256×256 gray-scale images, namely Synthetic, Barbara, Boat, Lena, and Fruits. Due to page restrictions, this paper presents only the visual assessment is presented here only for a synthetic image (depicted in Fig. 2), but the evaluation of the perfect reconstruction property is given for all experimental images. Fig. 2(a) shows the original synthetic image, whereas Fig. 2(b, c, d) and (e, f, g) shows the decomposed and reconstructed images via DWT, MR-SVD, and R-MR-SVD, respectively. For DWT, the Cohen–Daubechies–Feauveau (CDF) 9/7 wavelet is used to decompose the experimental images. In order to explore the perfect reconstruction property of these transforms, all the experimental images are decomposed by DWT, MR-SVD, and R-MR-SVD followed by the reconstruction process. Once the experimental images are reconstructed, the similarity is measured between original and reconstructed images. For this purpose, PSNR, UQI, and SSIM are chosen. The PSNR indicates the similarity between two images and is usually expressed in terms of the logarithmic decibel scale, whereas UQI and SSIM are the similarity measures based on the structural information present in the images. The mathematical definitions of these similarity measures are as follows:

1. Peak Signal to Noise Ratio (PSNR): In principle, the higher the value of PSNR, the greater the similarity in the images. Mathematically, PSNR is defined as

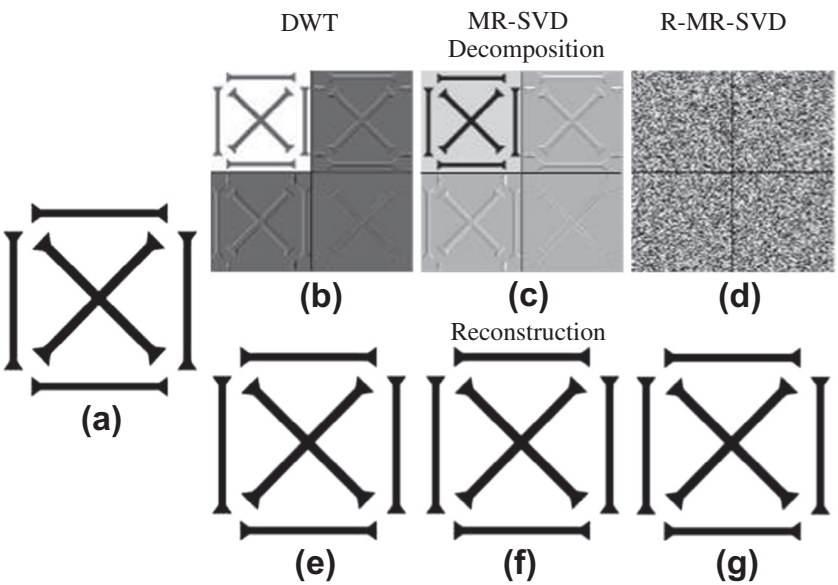

Fig. 2. Comparison between wavelet, multiresolution SVD and random multiresolution SVD. (a) Original image; decomposed image using (b) DWT, (c) MR-SVD, (d) R-MR-SVD; reconstructed image using (e) DWT, (f) MR-SVD and (g) R-MR-SVD.

$$\mathrm{PSNR}(x,y) = 10\log_{10}\frac{L^2}{\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}[x_{i,j}-y_{i,j}]^2} \tag{1}$$

where MN is the total number of pixels in the image, $x_{i,j}$ and $y_{i,j}$ are the values of the ijth pixel in the original and reconstructed image. In contrast, L is the maximum possible pixel value of the image, which is 255 when the pixels are represented using 8 bits per sample.

2. Universal Image Quality Index (UQI): The UQI indicates the structural similarity between two images. The UQI lies between $[-1,1]$ and the closer the value is to 1, the greater the similarity in the images. Mathematically, UQI is defined as

$$\mathrm{UQI}(x,y) = \frac{\sigma_{xy}}{\sigma_x\sigma_y}\cdot\frac{2\mu_x\mu_y}{\mu_x^2+\mu_y^2}\cdot\frac{2\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2} \tag{2}$$

where $\mu_x, \mu_y, \sigma_x, \sigma_y$ and σ_{xy} are the mean of x, y, variance x, y and the covariance of x, y respectively.

3. Structural Similarity Index Measure (SSIM): The SSIM is the extended version of the UQI index. The SSIM is also lies between $[-1,1]$ and the closer the value is to 1, the greater the similarity in the images. Mathematically, SSIM is defined as

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)} \tag{3}$$

where C_1, C_2 are two constants and are used to stabilize the division with a weak denominator.

PSNR is the most frequently used measure for signal fidelity. It essentially gives the ratio between the maximum possible power of a signal and the power of distortion that affects the fidelity of its representation. Due to the very wide dynamic range of signals, PSNR is usually expressed in terms of the logarithmic decibel scale. Therefore, a small difference value of the PSNR implies high numerical differences between the images. However, PSNR lacks the qualities for a perceptual image quality measure, as stated in [24]. Therefore, to check the perceptual quality of the images, UQI and SSIM are used. Unlike PSNR, UQI and SSIM take the advantage of the fact that the image degradation is a perceived change in structural information. Again it is pointed out in [24] that UQI can suffer from instability under certain conditions, for instance, in the case of division with a weak denominator. Therefore, SSIM is considered to corroborate the analysis. One of the appealing properties of SSIM is its boundedness, which signifies that any value less than 1 is due to the presence of perceptual quality degradation. In this section, these measures are used to check how faithful the reconstruction process is for the aforementioned MR transforms. In Section 5, these measures tell us how reliable the encryption process is in terms of signal fidelity. The values of these similarity measures are shown in Table 1 for all experimental images. It is clear from the values of PSNR that signal reconstruction is more accurately achieved by MR-SVD and R-MR-SVD compared to DWT. Also, the values of SSIM and UQI reveal that perfect perceptual quality of the reconstructed images can be achieved by MR-SVD and R-MR-SVD.

4.1. Analysis with varying values of p and q

In MR-SVD and R-MR-SVD, the values of p and q define the number of sub-bands obtained for a given image. According to the principle, the first row of matrix $\widehat{X}$ corresponds to the largest singular value, hence $\widehat{X}$ is considered the approximate part of the signal. Similarly, the remaining rows of $\widehat{X}$ correspond to the detail parts of the signal (see Sections 2 and 3 for more information about scatter and $\widehat{X}$ matrices). Since the size of matrix $\widehat{X}$ is $pq \times pq$, we obtain one approximate (corresponding to the first row) and $pq-1$ detail parts (corresponding to the remaining rows); for instance, if $p=q=2$, one approximate and three detail parts are obtained. After arranging these parts into a matrix, one can construct the MR-SVD/R-MR-SVD of the 2D signal X. Mathematically, $\phi_1 = \widehat{X}(1,:)$ and $\psi_1^\theta = \widehat{X}(\theta_i,:)$, where θ represents the orientation i.e. $\theta = H$ iff $\theta_i = 2, \theta = V$ iff $\theta_i = 3$ and $\theta = D$ iff $\theta_i = 4$. Hence, for $p=q=2$ we get one approximate and three detail parts as obtained in DWT. The MR-SVD/R-MR-SVD is a flexible and superior tool for signal processing and analysis, since one approximate and more than three detail parts are obtained whenever $p \neq q > 2$. The visual assessment of different sub-bands obtained by the different values of p and q is depicted in Fig. 3.

Table 1
Perfect reconstruction analysis using similarity measures.

Metric	DWT			MR-SVD			R-MR-SVD		
	PSNR (dB)	UQI	SSIM	PSNR (dB)	UQI	SSIM	PSNR (dB)	UQI	SSIM
Synthetic	213.0489	0.9961	0.9987	253.8244	1	1	253.8064	1	1
Barbara	216.1668	0.9960	0.9997	254.6953	1	1	254.4466	1	1
Boat	215.6561	0.9913	0.9968	251.1459	1	1	251.2185	1	1
Lena	215.2406	0.9904	0.9964	252.3671	1	1	251.3873	1	1
Fruits	215.0956	0.9912	0.9933	254.4062	1	1	254.0537	1	1

4.2. Analysis with energy compaction

If Fig. 2(b and c) are considered, one can observe that the compaction/information of energy in MR-SVD is better than that of DWT. To prove our claim, the entropies [9] of the coefficients obtained by DWT and MR-SVD are calculated and compared. Let us denote coefficients by $C_{i,j}^{l,\theta}$ where i and j denote the coefficient indices and l and θ denote the level of decomposition and orientation. The energy of the coefficients at each level and orientation is given by

$$E_{l,\theta} = \sum_i \sum_j |C_{i,j}^{l,\theta}|^2 \tag{4}$$

and the total energy is obtained as

$$E_{total} = \sum_l \sum_\theta \sum_i \sum_j |C_{i,j}^{l,\theta}|^2 = \sum_l \sum_\theta E_{l,\theta} \tag{5}$$

Now the relative energy of coefficients is given by

$$P_{l,\theta} = \frac{E_{l,\theta}}{E_{total}} \tag{6}$$

Based on these equations, the total entropy of the coefficients (at lth level $\mathcal{E}_l$ and total $\mathcal{E}$) is defined as

$$\mathcal{E}_l = -\sum_\theta P_{l,\theta} \quad \ln(P_{l,\theta}), \quad \mathcal{E} = \sum_l \mathcal{E}_l \tag{7}$$

The entropy of the coefficients appears as a measure of the degree of order/disorder (information) that the signal/image possesses. Generally, entropy is used to calculate the correlation between oscillating structures corresponding to different scales. With entropy in particular, self-similar processes can be detected. For instance, the multiresolution representation of a very ordered process, which could be conceptualized as a signal with a narrow band spectrum, will be greatly dissolved in one unique resolution level, i.e., all relative energies of the coefficients will be almost zero except for the level that includes the representative signal frequency, i.e., the approximate part. For this special level, the relative energy will be almost one and consequently, the total entropy will be either zero or very close to zero. Furthermore, the multiresolution representation of a very disordered process, which could be thought of as a signal with a wide band spectrum will have significant contributions from all frequency bands. Moreover, the relative energy will be almost equal for all resolution levels, and consequently, entropy will take maximum values. In general, the smaller entropy values exhibit better energy information/compaction in the coefficients. For this purpose, the experimental images are decomposed by 4 level DWT, MR-SVD and R-MR-SVD followed by the entropy calculation using Eq. (7). The entropies are depicted in Table 2 for all images. As seen in the table, it is clear that the entropy increases with DWT compared to MR-SVD. This means that the energy is more spread out and not compacted in DWT compared to MR-SVD. Furthermore, the entropies for R-MR-SVD are significantly greater than those of DWT and MR-SVD. Introducing randomness in the transform leads to white noise for the coefficients, so it is unsurprising that the entropy is much higher than with MR-SVD. Hence, the proposed R-MR-SVD can introduce randomness perfectly.

Any multiresolution transform decomposes a signal into different multi-resolution sub-bands where the lowest resolution sub-band (approximation sub-band) is the smallest sub-band and includes high amplitude approximation coefficients.

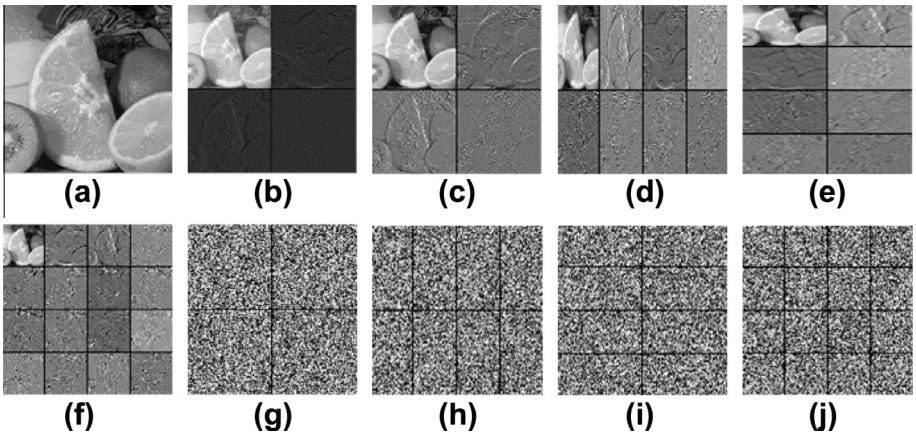

Fig. 3. Effect of different values of p and q on MR-SVD and R-MR-SVD. (a) Fruits image; decomposed image using (b) DWT, (c) MR-SVD with $p = 2, q = 2$, (d) MR-SVD with $p = 2, q = 4$, (e) MR-SVD with $p = 4, q = 2$, (f) MR-SVD with $p = 4, q = 4$, (g) R-MR-SVD with $p = 2, q = 2$, (h) R-MR-SVD with $p = 2, q = 4$, (i) R-MR-SVD with $p = 4, q = 2$ and (j) R-MR-SVD with $p = 4, q = 4$.

In contrast, the wavelet coefficients (detail coefficients), contained in the detail sub-bands, have small magnitudes. Therefore, most of the energy of the signal is captured by approximation coefficients allowing better energy compaction. Another way to prove the superior energy compaction in MR-SVD is to calculate the energy percentage contained in the sub-bands. In order to explore how much energy is contained in approximation and detail sub-bands, Energy Packing Efficiencies (EPE) are considered for all sub-bands. EPE for a set of coefficients in the sub-band is defined as the ratio of the energy captured by the sub-band coefficients and the energy captured by the whole number of coefficients. Mathematically, EPE is defined as

$$\mathrm{EPE}_\theta = \frac{E_{l,\theta}}{E_{total}} \times 100\% \tag{8}$$

where $E_{l,\theta}$ and E_{total} are given by Eqs. (4) and (5) respectively. Here $E_{l,\theta}$ is the energy of a sub-band at level l and orientation $\theta \in [A, H, V, D]$ whereas E_{total} is the total energy contained by all sub-bands. All experimental images are decomposed by 1-level DWT, MR-SVD and R-MR-SVD in order to see energy distribution. The values for p and q in MR-SVD and R-MR-SVD are taken to be 2, because these values give 4 sub-bands similar to DWT. The energy distribution is depicted in Table 3. It can be seen from the table that the energy compaction in MR-SVD is better than in DWT. For instance, if the synthetic image is considered, 97.31% of the total energy is concentrated in the approximation coefficients in DWT, whereas in MR-SVD this percentage is 98.51%.

4.3. Randomness analysis of R-MR-SVD

Due to the substantial proliferation of internet technologies, access to stored information on computer databases has increased greatly. These phenomenal developments lead to crucial problems, such as the availability, privacy, and integrity of information. In order to solve these problems, we rely on the information security. Information security is the process of protecting information and information systems from unauthorized access, use, disclosure, disruption, modification, perusal, inspection, recording, or destruction. Under current security measures, we see that encryption methodologies and cryptographic techniques have expanded in both complexity and usage. The application of these technologies is required to keep secrets safe and secure [4,5]. To that end, randomness plays a vital role in most cryptosystems, providing uniform randomness to the information so that information will become unintelligible and only accessible by authorized persons. 'Uniform randomness' means that the probability of selecting of any coefficient is almost the same. A brief discussion on this can be found in Section 4.3.1. As already validated experimentally through energy compaction, R-MR-SVD can provide randomness, see Section 4.2. Therefore, the foremost application of R-MR-SVD is in information security. To reinstate our claim, the randomness of the proposed R-MR-SVD is further analyzed. Here, randomness is generated by two random matrices, namely $\mathcal{K}_1$ and $\mathcal{K}_2$, created with initial seeds K_1 and K_2. The proposed framework uses a 1D chaotic map, namely the logistic map, to create random sequences [20]. The logistic map is chosen due to its attractive features such as extreme sensitivity to initial conditions, outspreading of orbits over the entire space, and its uncorrelated nature. Mathematically, the logistic map is defined as

$$z(i+1) = az(i)[1 - z(i)], \quad i = 1, 2, \ldots, \tag{9}$$

where $a \in (3.5699, 4]$, $z(i) \in (0, 1)$ and $z(1)$ is the initial seed. Keeping the chaotic parameters and the initial condition as the secret keys, the chaotic map can be reproduced easily, which further reduces the space complexity of the algorithm. After obtaining the sequence, the elements are stacked into matrix form (of the desired size) by the row-wise insertion strategy.

Table 2
Comparison of DWT, MR-SVD and R-MR-SVD for energy compaction based on entropy.

Image	Transform	Entropy				
Level		1	2	3	4	Total
Synthetic	DWT	0.2965	0.4283	0.1576	0.1609	1.0433
	MR-SVD	0.0910	0.1574	0.3534	0.4113	1.0131
	R-MR-SVD	0.5781	1.1720	1.1832	1.1505	3.6839
Barbara	DWT	0.1537	0.2391	0.2609	0.2455	0.8992
	MR-SVD	0.0835	0.1221	0.1752	0.1779	0.5588
	R-MR-SVD	0.2037	1.2500	1.1304	1.0933	3.6775
Boat	DWT	0.0986	0.1117	0.0947	0.1456	0.4506
	MR-SVD	0.0679	0.0836	0.0892	0.0900	0.3308
	R-MR-SVD	0.2019	1.3357	1.1035	1.2675	3.9086
Lena	DWT	0.1209	0.1231	0.1239	0.1925	0.5604
	MR-SVD	0.0602	0.0897	0.1093	0.1596	0.4189
	R-MR-SVD	0.1888	1.0639	0.9760	1.3523	3.5809
Fruits	DWT	0.1240	0.1023	0.1081	0.2329	0.5673
	MR-SVD	0.0375	0.0505	0.0762	0.1112	0.2753
	R-MR-SVD	0.2596	1.0743	1.2048	1.1120	3.5507

Table 3
Comparison of DWT, MR-SVD and R-MR-SVD for energy distribution based on EPE.

Image	Transform	EPE			
Orientation (θ)		A (%)	H (%)	V (%)	D (%)
Synthetic	DWT	97.31	0.69	1.36	0.64
	MR-SVD	98.51	0.70	0.66	0.12
	R-MR-SVD	96.55	0.37	2.40	0.68
Barbara	DWT	97.14	0.88	1.24	0.74
	MR-SVD	98.65	0.42	0.82	0.11
	R-MR-SVD	95.91	2.88	0.45	0.76
Boat	DWT	94.17	2.83	2.06	0.94
	MR-SVD	98.95	0.49	0.45	0.10
	R-MR-SVD	96.17	1.89	1.23	0.72
Lena	DWT	97.72	0.74	0.77	0.77
	MR-SVD	99.08	0.23	0.59	0.10
	R-MR-SVD	96.20	2.85	0.33	0.62
Fruits	DWT	98.44	0.77	0.27	0.52
	MR-SVD	99.48	0.25	0.19	0.08
	R-MR-SVD	97.02	0.93	1.82	0.22

In the proposed work, the value of a is taken to be 3.58 and two random sequences are generated using initial seeds $K_1 = 0.6991$ and $K_2 = 0.4560$ and stacked in the matrix form of size $pq \times pq$ and $mn/pq \times mn/pq$.

In order to show sensitivity, the keys are changed slightly and then the image is reconstructed by inverse R-MR-SVD. The modified values of the keys are taken as $K_1 = 0.6991005$ and $K_2 = 0.45600009$ and are approximately the same as the original values. The results of sensitivity are depicted in Fig. 4. Fig. 4(a) shows the original Fruits image whereas Fig. 4(b and c) shows the 1 level MR-SVD and R-MR-SVD transformed images. The reconstructed image with correct keys is shown in Fig. 4(d). Fig. 4(e) shows the reconstructed image with the wrong random matrix $\mathcal{K}_1$ (obtained by initial seed $K_1 = 0.6991005$), whereas Fig. 4(f) shows the reconstructed image with the wrong random matrix $\mathcal{K}_2$ (obtained by initial seed $K_2 = 0.45600009$). Fig. 4(g and h) shows the results when both random matrices are wrong. As seen in the figures, it is clear that if any random matrix is wrong, perfect reconstruction cannot be achieved. Hence, the proposed R-MR-SVD is highly sensitive to the random matrices, i.e., to the initial seeds.

To review the randomness analysis for the R-MR-SVD, the histogram, the correlation plot between adjacent pixels, and the relative randomness to the MR-SVD are considered in the following sections.

4.3.1. Histogram analysis

The histogram analysis is undertaken to compare the histograms of the sub-bands obtained by MR-SVD and R-MR-SVD. In principle, the color histograms of the sub-bands should be uniformly distributed in order to appear random. For a good level

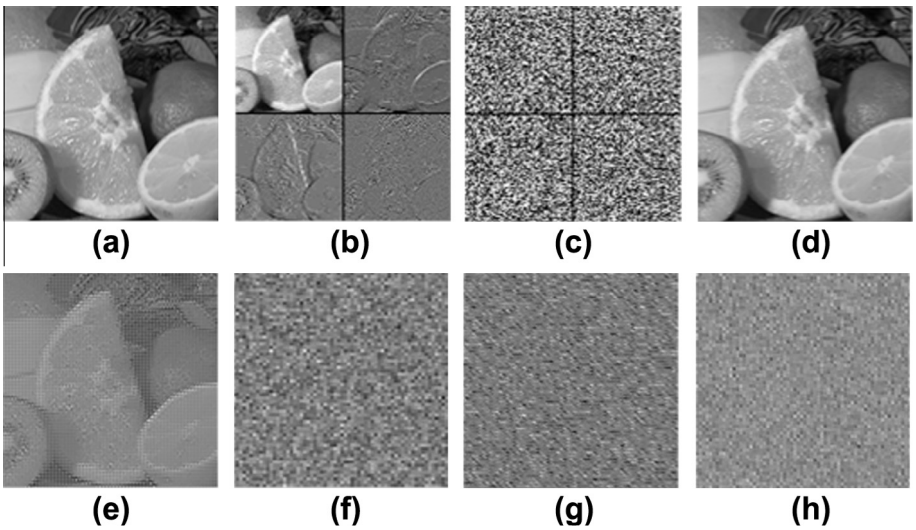

Fig. 4. Key sensitivity of R-MR-SVD. (a) Original image; decomposed image using (b) MR-SVD, (c) R-MR-SVD using $K_1 = 0.6991$ and $K_2 = 0.4560$; reconstructed image using (d) all correct keys ($K_1 = 0.6991$ and $K_2 = 0.4560$), (e) wrong key K_1 ($K_1 = 0.6991005$), (f) wrong key K_2 ($K_2 = 0.45600009$), (g) wrong keys K_1, K_2 ($K_1 = 0.6991005$ and $K_2 = 0.45600009$) and (h) wrong keys K_1, K_2 ($K_1 = 0.2398$ and $K_2 = 0.0125$).

of randomness, the histograms of the sub-bands must be fairly uniformly distributed. For this experiment, the Fruits image is decomposed first by the 1 level MR-SVD, then by the 1 level R-MR-SVD, followed by the histogram plotting. The histograms for all the sub-bands are depicted in Fig. 5(b–i), which clearly demonstrate that the histograms of R-MR-SVD sub-bands are fairly uniform. If the histogram of the original image is compared with the histogram of approximate sub-band of MR-SVD, one can observe that there is a similarity between them, further demonstrating that the approximate sub-band is the smoother version of original image. In contrast, the histograms of the R-MR-SVD sub-bands as shown in Fig. 5(f–i) have uniform distribution, which is significantly different from that of the MR-SVD sub-bands; i.e., there are no statistical similarities between the MR-SVD and R-MR-SVD sub-bands. In other words, the original image histogram has four different modes (Fig. 5(a)). As expected, after applying MR-SVD, the histogram of approximate band has four modes (Fig. 5(b)) that are similar to the original histogram. Conversely, the histograms of different detail bands (Fig. 5(c–e)) have single mode, which shows that there is no significant randomness in the detail bands. However, in Fig. 5(f–i), the histograms of the R-MR-SVD bands do not have distinct modes and roughly resemble uniform distribution. This explains that the probability of selecting of any coefficient is almost the same. Throughout this paper, this phenomenon is termed uniform randomness. The uniformity is also further justified by the chi-square test [17] as follows.

$$\chi^2_{test} = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{e_i} \tag{10}$$

where k is the number of levels with o_i and e_i are the observed and expected occurrence frequencies of each level respectively. With a significance level of 0.05, $\chi^2_{test} < \chi^2_{255,0.05}$ is found for R-MR-SVD sub-bands, which implies that the null hypothesis cannot be rejected and the distribution of the histogram is uniform. Conversely, for MR-SVD sub-bands, $\chi^2_{test} > \chi^2_{255,0.05}$, which implies that the null hypothesis is rejected and the distribution of the histogram is not uniform. Therefore, there is a superior level of randomness in R-MR-SVD sub-bands compared to MR-SVD sub-bands.

4.3.2. Correlation analysis

For a natural image, each pixel is usually highly correlated to adjacent pixels. This correlation can be horizontal, vertical, diagonal, or in all directions. Similarly, the correlation between adjacent coefficients of the sub-bands should be close to zero to appear random. The following procedure was carried out to test the correlation between vertically, horizontally, or diagonally adjacent pixels. Select any pixel at any random location and form a vector considering its adjacent pixel in a particular direction. Similarly, form P vectors corresponding to more random locations. Then calculate the correlation coefficient between each pair of vectors using the following formula:

$$r_{xy} = \frac{cov(x,y)}{\sqrt{var(x)}\sqrt{var(y)}} \tag{11}$$

with $cov(x,y) = E(x - E(x))(y - E(y))$, $var(x) = \frac{1}{P}\sum_{i=1}^{P}(x - \hat{x})^2$, $var(y) = \frac{1}{P}\sum_{i=1}^{P}(y - \hat{y})^2$, where x and y are gray levels of two adjacent coefficients in the sub-bands. Fig. 5(k–r) shows the correlation distribution of two horizontally adjacent coefficients in the MR-SVD and R-MR-SVD sub-bands. The similarity between the correlation plots of the approximate sub-band of MR-SVD and of the original image proves that the approximate sub-band is the smoother version of the original image. As the detail sub-bands contain the texture and edges present in the images, high correlation is obtained in these sub-bands as well. The correlation coefficients for MR-SVD and R-MR-SVD sub-bands in all directions are shown in Table 4. As shown in Table 4, the R-MR-SVD coefficients are random since they have a correlation close to zero, whereas MR-SVD adjacent coefficients have a very good correlation close to one. Therefore, there exists more randomness in R-MR-SVD sub-bands compared to MR-SVD sub-bands.

5. Application of R-MR-SVD to image encryption

Due to the substantial proliferation in communication and information technologies, the security of digital transmission is becoming increasingly important. In most cases, the transmission channels are not secure enough to prevent illegal access by malicious intruders. Recently, information security is identified as the possible solution for these scenarios. Information security is the process of protecting information and information systems from counterfeiting, tampering, and unauthorized access. To meet these challenges, robust encryption (cryptographic) techniques are of vital importance. Encryption techniques employ a kind of randomness that cannot be inferred by other unauthorized users. Therefore, randomness plays a vital role in most cryptosystems [6,7]. The aim of encryption techniques is to give uniform randomness to the information so that information becomes unintelligible to intruders and only accessible by authorized persons.

The existing multiresolution representations, including DWT and MR-SVD, have no concepts on randomness. This limits applicability and usability in information security [7]. From the earlier discussion, clearly R-MR-SVD is a realization of intrinsic randomness with multiresolution representation. Therefore, R-MR-SVD can play a remarkable role in information security. Hence, when an application needs randomness as its key factor, R-MR-SVD is expected to provide better results. To support our claim, an encryption technique is proposed for images. Images have ample applications in our daily lives and are the basis of most security verification systems, which are widely used for the identification of people and other entities.

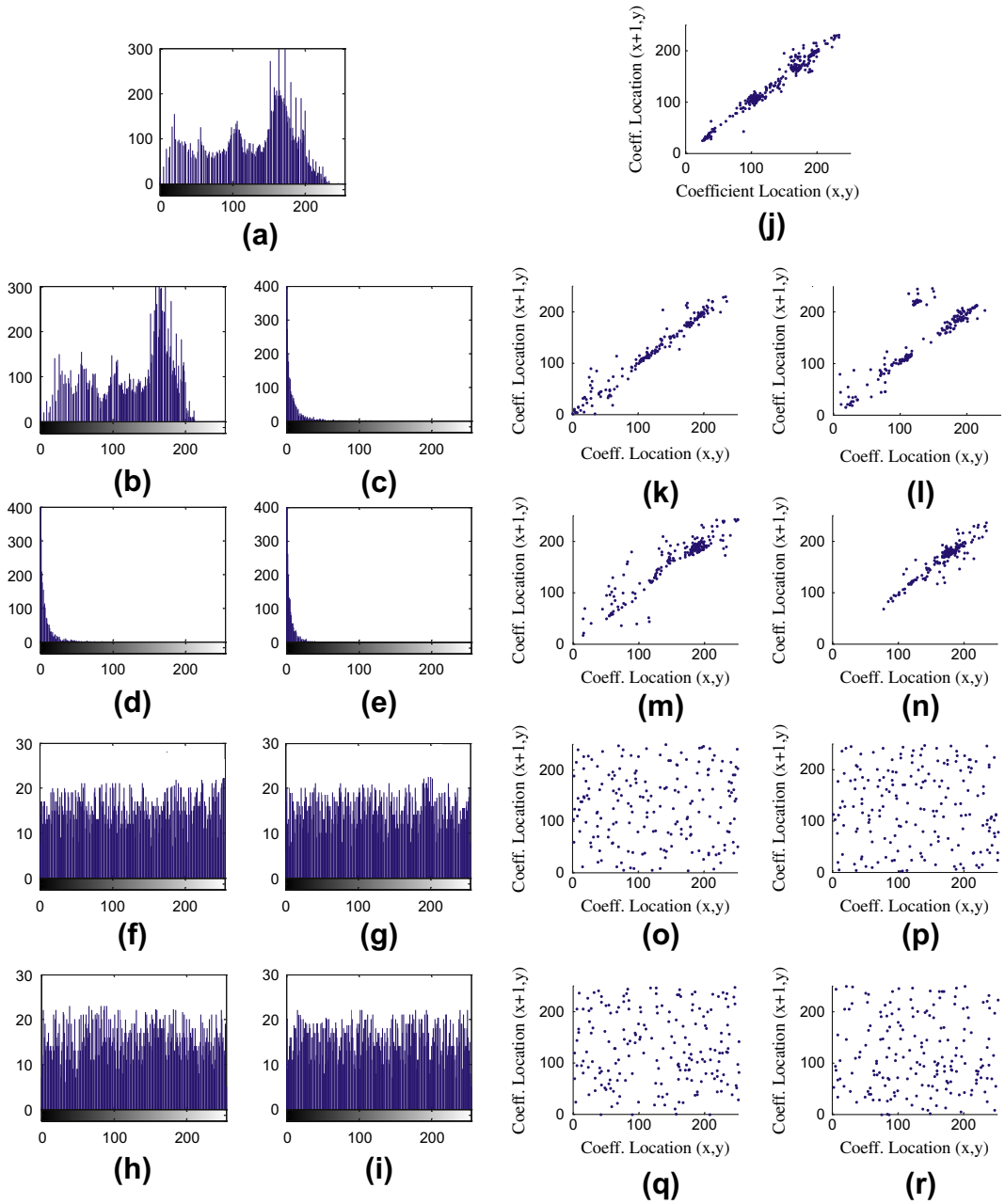

Fig. 5. Histogram and correlation plot of (a, j) fruit image, (b, k) approximate sub-band of MR-SVD, (c, l) horizontal sub-band of MR-SVD, (d, m) vertical sub-band of MR-SVD, (e, n) diagonal sub-band of MR-SVD, (f, o) approximate sub-band of R-MR-SVD, (g, p) horizontal sub-band of R-MR-SVD, (h, q) vertical sub-band of R-MR-SVD and (i, r) diagonal sub-band of R-MR-SVD.

In this section, an image encryption technique is proposed to demonstrate the efficiency of R-MR-SVD in information security. Without the loss of generality, we choose a very simple encryption and decryption processes that perfectly demonstrate the superiority of R-MR-SVD over MR-SVD and DWT for security purposes. Assume that F represents the original image of size $M \times N$. The proposed encryption technique is depicted in Fig. 6 and can be summarized as follows:

1. Perform ℓ-level R-MR-SVD on the original image F, which is denoted by f_l^θ, where $\theta \in \{A, H, V, D\}$ is the orientation of the sub-bands and $l \in [1, \ell]$ is the corresponding level.

Table 4
Comparison of MR-SVD and R-MR-SVD for correlation between adjacent coefficients in all directions.

Sub-bands	Image	Transform	Correlation between adjacent coefficients in		
			Horizontal direction	Vertical direction	Diagonal direction
Approximation sub-band	Synthetic	MR-SVD	0.9806	0.9124	0.8853
		R-MR-SVD	−0.0088	0.0026	−0.0035
	Barbara	MR-SVD	0.9813	0.9180	0.8817
		R-MR-SVD	−0.0033	0.0030	0.0036
	Boat	MR-SVD	0.9798	0.9157	0.8847
		R-MR-SVD	−0.0084	−0.0187	−0.0127
	Lena	MR-SVD	0.9837	0.9068	0.8975
		R-MR-SVD	−0.0040	0.0103	−0.0727
	Fruits	MR-SVD	0.9793	0.9262	0.8798
		R-MR-SVD	0.0016	0.0121	−0.0196
Horizontal sub-band	Synthetic	MR-SVD	0.8535	0.8504	0.8118
		R-MR-SVD	0.0194	0.0134	0.0174
	Barbara	MR-SVD	0.8576	0.8353	0.8151
		R-MR-SVD	0.0103	0.0045	0.0245
	Boat	MR-SVD	0.8489	0.8370	0.8228
		R-MR-SVD	0.0240	0.0142	−0.0064
	Lena	MR-SVD	0.8558	0.8211	0.8090
		R-MR-SVD	0.0122	−0.0042	0.0041
	Fruits	MR-SVD	0.8498	0.8210	0.8284
		R-MR-SVD	−0.0033	−0.0021	0.0112
Vertical sub-band	Synthetic	MR-SVD	0.8560	0.8228	0.8351
		R-MR-SVD	−0.0091	0.0171	0.0010
	Barbara	MR-SVD	0.8330	0.8341	0.8078
		R-MR-SVD	−0.0074	0.0228	0.0121
	Boat	MR-SVD	0.8550	0.8246	0.8341
		R-MR-SVD	0.0001	0.0078	0.0275
	Lena	MR-SVD	0.8454	0.8318	0.8007
		R-MR-SVD	0.0091	0.0147	0.0068
	Fruits	MR-SVD	0.8468	0.8452	0.8202
		R-MR-SVD	0.0339	0.0013	−0.0051
Diagonal sub-band	Synthetic	MR-SVD	0.9370	0.9066	0.8466
		R-MR-SVD	−0.0191	0.0032	−0.0015
	Barbara	MR-SVD	0.9243	0.8969	0.8417
		R-MR-SVD	0.0028	0.0091	0.0083
	Boat	MR-SVD	0.9360	0.9071	0.8459
		R-MR-SVD	0.0043	0.0101	0.0031
	Lena	MR-SVD	0.9414	0.9062	0.8517
		R-MR-SVD	−0.0005	0.0114	0.0038
	Fruits	MR-SVD	0.9283	0.9097	0.8388
		R-MR-SVD	−0.0019	−0.0012	0.0005

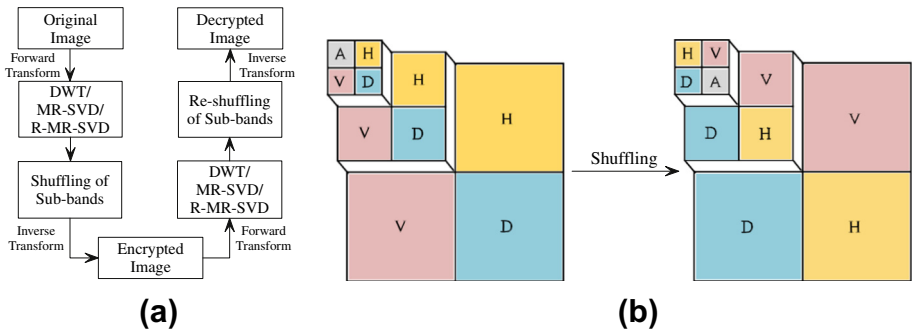

Fig. 6. (a) Block diagram of proposed encryption technique and (b) shuffling of sub-bands in 3-level decomposition.

2. Shuffle the orientation of the sub-bands from the coarsest to the finest level. For this purpose, the positions of the sub-bands are changed from $\{A, H, V, D\}$ to $\{D, V, H, A\}$ (see Fig. 6(b) for shuffling of sub-bands in 3-level decomposition).
3. Perform inverse ℓ-level R-MR-SVD on the shuffled sub-bands to get the encrypted image ($\mathcal{F}$).

Image decryption is an inverse process of encryption and can be formulated as follows.

1. Perform ℓ-level R-MR-SVD on the encrypted image $\mathcal{F}$, which is denoted by $\widetilde{f}_l^{\theta}$, where θ is the orientation of the sub-bands and l is the corresponding level.
2. Re-shuffle the orientations of the sub-bands from the coarsest to the finest level to their original positions. For this purpose, the positions of the sub-bands are changed from $\{D, V, H, A\}$ to $\{A, H, V, D\}$.
3. Perform inverse ℓ-level R-MR-SVD on the re-shuffled sub-bands to get the decrypted image ($\widetilde{F}$).

To demonstrate the superior performance of the proposed scheme, we complete a more elaborate performance comparison with DWT and MR-SVD. The key idea is to replace R-MR-SVD with DWT and MR-SVD in the aforementioned encryption-decryption process. The performance of the proposed encryption process is established using the MATLAB platform considering synthetic and Lena images. The encrypted and decrypted images are shown in Figs. 7 and 8 respectively. These figures show the encrypted and decrypted images with DWT, MR-SVD and R-MR-SVD respectively. The encrypted images from DWT and MR-SVD give the overview of the original image, which violates the foremost cryptosystem requirement of producing an unintelligible image. Therefore, DWT and MR-SVD cannot be used for security purposes unless a diffusion process is applied on the coefficients. If the encrypted images from DWT and MR-SVD are compared, it is clear that the performance of MR-SVD is better than that of DWT. This is due to the superior energy compaction of MR-SVD. As the sub-band positions change, the energy is distributed irregularly over the image when inverse transform is applied; therefore, the encrypted image is virtually unintelligible. Conversely, one cannot get the overview of the original image from the encrypted images obtained from R-MR-SVD, i.e., the encrypted images are perfectly unintelligible. Furthermore, Fig. 8 shows the decrypted images with DWT, MR-SVD and R-MR-SVD respectively. Clearly, all three transforms have performed equally. As proved in Section 4.3, R-MR-SVD is highly sensitive to the keys. Therefore, using R-MR-SVD introduces the key concept in the proposed encryption framework. Thus, R-MR-SVD plays a remarkable role in information security.

Security is the main goal of encryption techniques. Some security analyses have been performed on the proposed encryption process that include spectrum analysis, edge similarity analysis, information entropy analysis and numerical analysis. These security analyses are discussed in the following subsections.

5.1. Spectrum analysis

In encryption, the basic idea is to compare the amplitude spectrum of the original and encrypted media to judge the effectiveness of the encryption. In principle, if the amplitude spectrum of the encrypted image is fairly uniform and is significantly different from that of the original image, the encryption is said to be perfect. The corresponding amplitude spectra of the original and encrypted images are depicted in Fig. 9. Fig. 9(a and e) shows the peak in the middle, which corresponds to the highest narrow spectra. In essence, the frequency distribution of the original image is concentrated in a small area in the middle, which usually runs the risk of information leakage during transmission. Hence, if the frequency distribution is concentrated over the entire image, the encryption is said to be perfect, since the risk of information leakage is minimal. Fig. 9(d and h) demonstrates that the spectra of encrypted images are uniform and prove the prevention of information

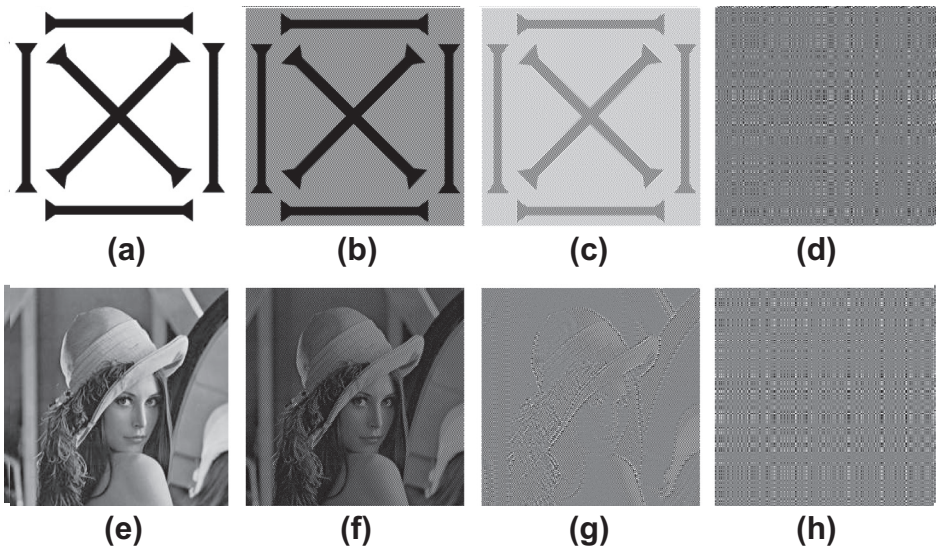

Fig. 7. Results for encryption process: (a, e) original images; encrypted images using (b, f) DWT, (c, g) MR-SVD and (d, h) R-MR-SVD.

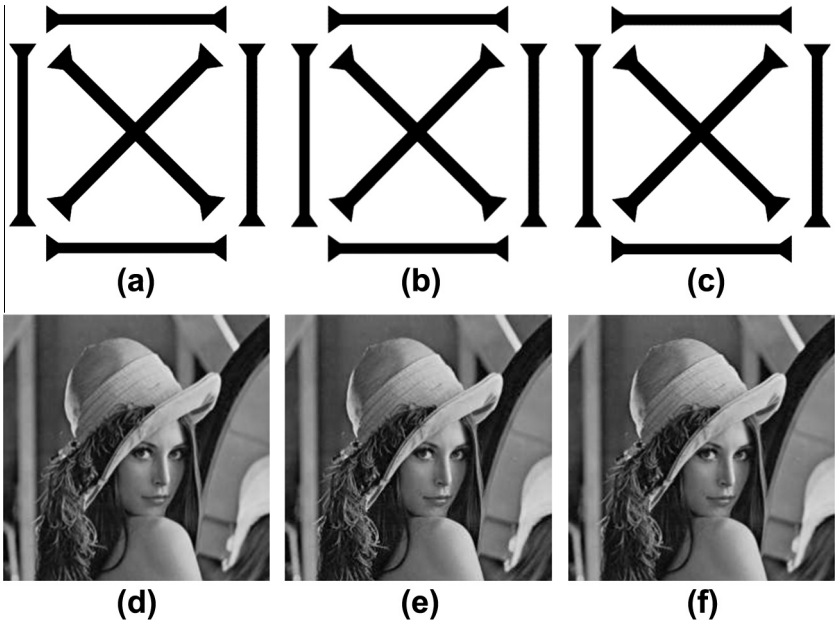

Fig. 8. Decrypted images using (a, d) DWT, (b, e) MR-SVD and (c, f) R-MR-SVD.

leakage. Furthermore, Fig. 9(b and f) and (c and g) shows the amplitude spectra of the encrypted images with DWT and MR-SVD, which are similar to the original. Therefore, unlike R-MR-SVD, the use of DWT and MR-SVD caused information leakage and the original image is not hidden perfectly against a statistical attack.

5.2. Edge similarity analysis

Edges are the most common features of an image. They are local variations in the image function. Typically, edges reflect the inherent properties of the objects (such as surface markings and surface shape) and the geometry of the scene. For a good encryption technique, these properties might not have been revealed from their encrypted counterparts. For this purpose, similar edges are estimated using the edge similarity ratio (ESR). Let f and g denote original and corresponding encrypted images with their respective binary edge matrices B_f and B_g. The mathematical definition of ESR is given as follows.

$$\mathrm{ESR} = \frac{\sum_{i=1}^{M}\sum_{i=1}^{N}|B_f(i,j) - B_g(i,j)|}{\sum_{i=1}^{M}\sum_{i=1}^{N}\left(B_f(i,j) + B_g(i,j)\right)} \times 100\% \tag{12}$$

Here, the edge binary matrix is obtained using the standard Canny edge detector. Higher values of ESR show that there is less similarity between the edges. Therefore, higher values of ESR are required for a good encryption technique. The values of ESR for experimental images are given in Table 5. As shown in the table, more than 99% of the edges in the original image are

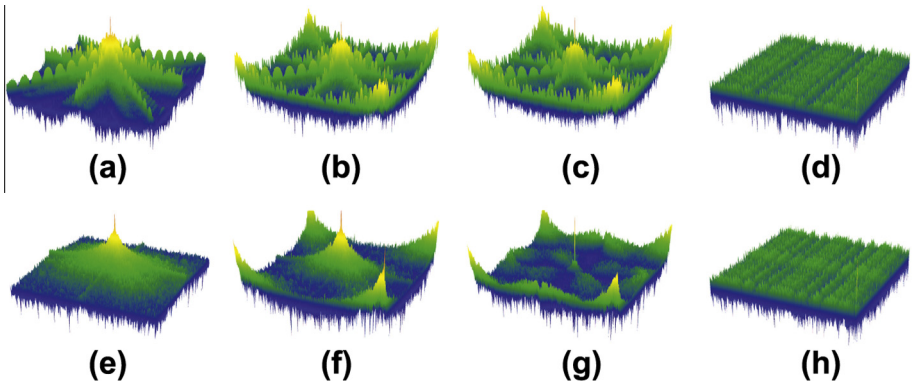

Fig. 9. The spectrum analysis for the proposed technique: amplitude spectra of synthetic and Lena image for (a, e) original image; encrypted image with (b, f) DWT, (c, g) MR-SVD and (d, h) R-MR-SVD.

Table 5
Edge similarity and information entropy analysis of proposed encryption technique.

Metric	Image	Encryption with		
		DWT	MR-SVD	R-MR-SVD
Edge similarity ratio	Synthetic	3.0558	22.3973	99.0402
	Lena	14.1197	70.3328	99.6157
Information entropy	Synthetic	4.8147	5.9134	7.9975
	Lena	4.9058	6.6324	7.9995

Table 6
Numerical analysis of proposed encryption technique.

Image	Metric	Encryption			Decryption		
		PSNR	UQI	SSIM	PSNR	UQI	SSIM
Synthetic	DWT	11.0732	0.1584	0.1759	246.5751	1	1
	MR-SVD	7.7736	0.0782	0.0686	246.7059	1	1
	R-MR-SVD	4.8261	0.0001	0.0026	246.6489	1	1
Lena	DWT	12.2741	0.1578	0.1708	249.3399	1	1
	MR-SVD	9.5384	0.0704	0.0643	249.5917	1	1
	R-MR-SVD	4.2219	0.0003	0.0059	249.5749	1	1

displaced from their positions in the encrypted image when R-MR-SVD is used whereas 14% and 70% of the edges are displaced with DWT and MR-SVD respectively. Hence, the inherent properties and geometry of the images cannot be revealed from the proposed R-MR-SVD based encryption framework.

5.3. Information entropy analysis

Information entropy expresses the degree of uncertainty (randomness) in the system. Therefore, it is considered one of the most important features of randomness. The information entropy $H(s)$ of a source s with 2^N symbols s_i is defined as

$$H(s) = -\sum_{i=1}^{2^N-1} \mathbb{P}(s_i)\log_2 \mathbb{P}(s_i) \tag{13}$$

where $\mathbb{P}$ denotes the probability of the symbol s_i being emitted from s. If s is a truly random source and $\mathbb{P}(s_i) = 2^{-N}, \forall i$, we have, $H(s) = 2^N 2^{-N} \log_2 2^N = N$. Therefore, the information entropy must be 8 for a random source emitting 256 symbols. For the encrypted images, the information entropies are listed in Table 5. For DWT and MR-SVD, the entropies are lower, however they are very close to the ideal value of 8 for R-MR-SVD. This means that the formed encrypted images using R-MR-SVD are close to a random source and are thus secure against entropy attack.

5.4. Numerical analysis

Finally, numerical analysis evaluates the proposed framework. Numerical analysis includes the values of the objective metrics. PSNR, UQI, and SSIM are used as the objective metrics to evaluate the proposed technique (mathematical definitions can be seen in Section 4). Table 6 shows the values of the objective metrics between the original-encrypted and original-decrypted images, for all experimental images. As shown in Table 6, R-MR-SVD outperforms DWT and MR-SVD in the encryption process, whereas all transforms perform equally in the decryption process. Therefore, R-MR-SVD must be chosen over other multiresolution representations for security purposes.

6. Application of R-MR-SVD to image compression

Although the main contribution of this paper is the introduction of R-MR-SVD for information security, an application of R-MR-SVD to lossy gray-scale image compression is presented in this section. The proposed R-MR-SVD is compared to DWT and MR-SVD in terms of the level of compression and compressed image quality. In this section, the EZW compression algorithm, which uses the wavelet transform is used to prove its effect on lossy image compression [18]. The EZW algorithm states that if a coefficient at a certain level of decomposition is less than a threshold (T), then all the coefficients of the same orientation in the same spatial location at lower decomposition levels are insignificant compared to T. In principle, if a coefficient is larger than T, it is encoded and removed from the image; if it is smaller, it is left for the next pass. When all coefficients have been visited, the threshold is lowered and the image is scanned again to add more detail to the already encoded

image. This process is repeated until all the wavelet coefficients have been encoded completely, or until another criterion has been satisfied.

In this study, we have replaced wavelet transform by MR-SVD and R-MR-SVD to judge the effect of these transforms on lossy image compression. See Fig. 10 for an enhanced overview. Fig. 11 shows the coded synthetic and Lena images for the

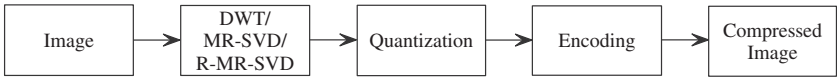

Fig. 10. Block diagram of image compression process.

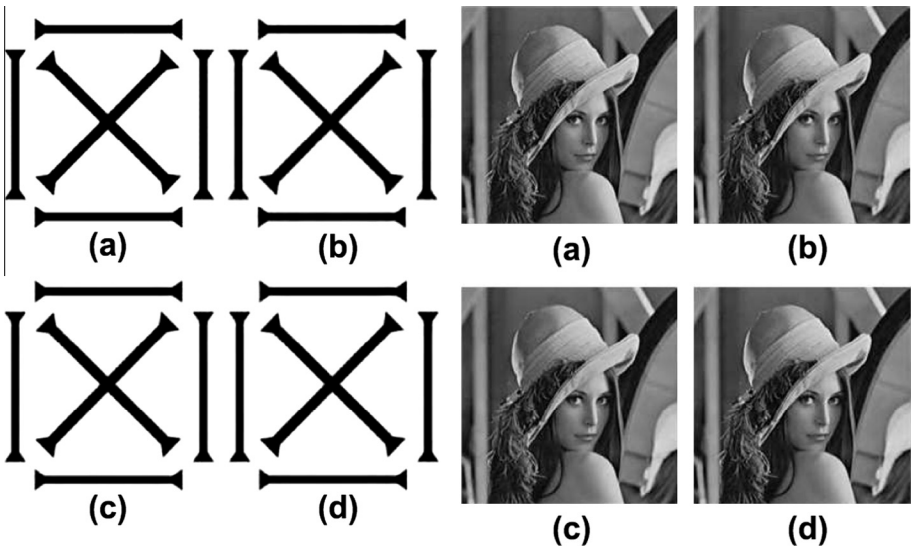

Fig. 11. Coded synthetic and Lena images with CR = 20: (a) original images; coded images with (b) DWT, (c) MR-SVD and (d) R-MR-SVD.

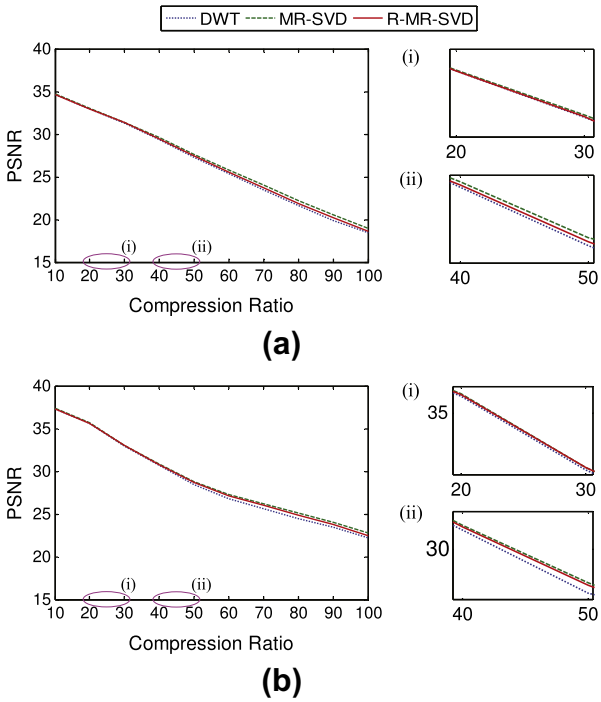

Fig. 12. Compression ratio-distortion curves for DWT, MR-SVD and R-MR-SVD, for both test images depicting: (a) synthetic and (b) Lena.

EZW compression algorithm with DWT, MR-SVD and R-MR-SVD respectively. Conversely, Fig. 12 gives compression ratio-distortion curves where distortion is measured in terms of PSNR. Results prove that the performance of MR-SVD is slightly better than that of DWT whereas the R-MR-SVD can achieve the same or better image quality at the same level of compression. In general, in this compression setup, MR-SVD and R-MR-SVD achieve better image quality than DWT for high compression levels, whereas for lower compression levels, they have the same image quality. For example, when the compression ratio is 70, MR-SVD achieves a PSNR improvement of 0.4 db over DWT for the synthetic image and 0.9 db for the Lena image. Alternatively, R-MR-SVD achieves a PSNR improvement of 0.3 db over DWT for the synthetic image and 0.75 db for the Lena image. Therefore, all three transforms have almost the same performance at lower compression ratios, but a significant difference comes at higher compression ratios.

7. Application of R-MR-SVD to face recognition

Another application of the proposed R-MR-SVD is in the field of face recognition. Multiresolution based face recognition has gained attention in the last few years [8] with the proliferation of multiresolution methods. Here, we demonstrate a comparison of R-MR-SVD-, MR-SVD-, and DWT-based face recognition using the Yale face database [27]. The database consists of 11 images of 15 subjects (165 images in all) with various facial expressions and configurations like center-light, with glasses, happy, left-light, without glasses, normal, right-light, sad, sleepy, surprised, and wink. In multiresolution-based face recognition, instead of pixel intensities, the multiresolution coefficients obtained across several scales are considered for forming the features of the face. In this comparison, we have used multiresolution coefficients directly to form features and have applied dimension reduction techniques on these coefficients to form feature vectors. For all experiments, for each subject, 7 images are used in training and the remaining 4 images are used for testing. Recognition rates reported are the average accuracy rates obtained by testing with three different combinations of training and test sets. A simple k-NN classifier with $k = 1$ is used for classification. Table 7 shows several comparisons based on R-MR-SVD, MR-SVD and DWT. In the first experiment, only the approximate sub-band is chosen. From the recognition rates, it is clear that dimension reduction by PCA followed by LDA gives the best result for each of the multiresolution decompositions. The performance of MR-SVD is the best, and R-MRSVD performs better than DWT. A similar trend in performances is observed for all sub-band based recognition. In all of the experiments, the performance of R-MR-SVD and MR-SVD is significantly better than that of DWT. Thus, it is experimentally validated that R-MR-SVD and MR-SVD can be used to form effective features for face recognition.

8. Discussion and future scope

First, we discuss the use of existing random transforms. There are two random transforms in the literature, namely, random singular value decomposition (R-SVD) [25] and random wavelet transform (R-WT) [26]. These two transforms are not included in the experimental analysis of the paper for the following reasons. R-SVD uses a randomized approach to calculate the singular value decomposition of a matrix. However, the output is the same for R-SVD and conventional SVD. The only difference comes in terms of the computational complexity, which is quite low in the case of R-SVD. On the contrary, R-WT is an attempt to decompose a signal with respect to a random filter rather than conventional wavelet filters. The main drawback of R-WT is that the use of random filters violates the perfect reconstruction condition associated with wavelet transform, and this results in an imperfect reconstruction process. Thus, the applicability of R-WT is restricted and is unsuitable for applications where the reconstructed process is necessary, as in the applications tested. As such, R-SVD and R-WT have not been considered in the experimental analysis of the paper.

In the earlier sections, a new variant of MR-SVD namely R-MR-SVD, is been proposed, and its utility has been demonstrated by applying it to three different applications. These applications include image encryption, image compression, and face recognition. The chosen applications are of significant importance in their respective domains under the broad field of image processing. Hence, the corresponding frameworks used are intended to give a clearer view of the better performance of R-MR-SVD over DWT and MR-SVD in these three applications. We do not intend to show or claim that the use

Table 7
Comparison of R-MR-SVD, MR-SVD and DWT for face recognition in Yale face database in terms of recognition rate (%).

Methods	Approximate sub-band	All sub-bands
R-MR-SVD	81.67	82.78
MR-SVD	81.67	83.89
DWT	77.78	78.33
R-MR-SVD + PCA	78.33	80.00
MR-SVD + PCA	78.89	80.00
DWT + PCA	73.33	74.44
R-MR-SVD + PCA + LDA	83.89	82.78
MR-SVD + PCA + LDA	85.00	84.44
DWT + PCA + LDA	80.00	80.00

of R-MR-SVD is superior to the traditional methods for in these three applications. However, it is worth considering how the performance of R-MR-SVD can be prepared to other relevant applications. Thus, future research should explore the use of R-MR-SVD in applications apart from those analyzed above. In the aforementioned applications, experimental results show better performance of R-MR-SVD over DWT and MR-SVD (refer to Sections 5–7). Further, design and implementation of efficient algorithms using R-MR-SVD is broader in scope and can be addressed separately in future research; this also comes in the future scope the proposed R-MR-SVD.

9. Conclusions

In this paper, MR-SVD is analyzed and generalized to R-MR-SVD by introducing intrinsic randomness. The proposed transform acquires the mathematical features of MR-SVD and possesses some unique features of its own. The main attributes of the new transform are the sensitivity of the initial keys and the randomness, which are further justified by the sensitivity and randomness analyses respectively. Therefore, R-MR-SVD can be used for a variety of applications such as information security, signal and image processing, and many more. In this work, three different applications, namely image encryption, lossy image compression, and face recognition, are considered to validate the practicality of R-MR-SVD. The first application shows the effectiveness of R-MR-SVD in information security and the other two show the same in image processing. The experimental results demonstrate the superior performance of R-MR-SVD over MR-SVD and DWT in the aforementioned applications. Still, questions remain concerning further aspects of R-MR-SVD and will be addressed in future research.

Acknowledgments

The work is supported in part by the Natural Sciences and Engineering Research Council (NSERC) of Canada under grant numbers 850007 and 371714.

Furthermore, the authors thank the anonymous Referees and the Editor for their valuable suggestions and many constructive comments that resulted in the improvement and readability of this paper.

References

[1] B. Akhbari, S. Ghaemmaghami, Watermarking of still images using multiresolution singular value decomposition, in: Proc. of Int. Sym. Intelligent Signal Processing and Communication Systems, IEEE, Hong Kong, 2005, pp. 325–328.
[2] R. Ashino, A. Morimoto, M. Nagase, R. Vaillancourt, Comparing multiresolution SVD with other methods for image compression, in: Proc. of Fourth ISAAC Congress, York University, Toronto, Canada, 2003, pp. 457–470.
[3] R. Ashin, A. Morimoto, R. Vaillancourt, Image compression with multiresolution singular value decomposition and other methods, Math. Comput. Model. 41 (6–7) (2005) 773–790.
[4] G. Bhatnagar, Q.M.J. Wu, Selective image encryption based on pixels of interest and singular value decomposition, Digital Signal Process. 22 (4) (2012) 648–663.
[5] G. Bhatnagar, Q.M.J. Wu, B. Raman, Image and video encryption based on dual space filling curves, Comput. J. 55 (6) (2012) 667–685.
[6] G. Bhatnagar, Q.M.J. Wu, Security solution for fingerprint data during communication and transmission, IEEE Trans. Instrum. Meas. 61 (4) (2012) 876–887.
[7] G. Bhatnagar, Q.M.J. Wu, B. Raman, A new fractional random wavelet transform for fingerprint security, IEEE Trans. SMC Part-A 42 (1) (2012) 262–275.
[8] H.K. Ekenel, B. Sankur, Multiresolution face recognition, Image Vision Comput. 23 (5) (2005) 469–477.
[9] M. Gell-Mann, C. Tsallis, Nonextensive Entropy: Interdisciplinary Applications, Oxford University Press, USA, 2004.
[10] K. Gerlach, S.D. Blunt, The factored-SVD formulation and an application example, Digital Signal Process. 17 (1) (2007) 199–208.
[11] G.H. Golub, C. Reinsch, Singular value decomposition and least squares solutions, Numer. Math. 14 (5) (1970) 403–420.
[12] R. Kakarala, P.O. Ogunbona, Signal analysis using a multiresolution form of the singular value decomposition, IEEE Trans. Image Process. 10 (5) (2001) 724–735.
[13] Y. Li, H. Zhang, Two properties of SVD and its application in data hiding, in: Proc. Int. Conf. on Advanced intelligent Computing theories and Applications, Springer, Qingdao, China, 2007, pp. 679–689.
[14] S.Y. Lung, Multi-resolution form of SVD for text-independent speaker recognition, Pattern Recognit. 35 (7) (2002) 1637–1639.
[15] D. Mukherjee, G. Bhatnagar, Q.M.J. Wu, Performance evaluation of multiresolution methods in disparity estimation, in: Proc. Int. Conf. Image and Signal Processing, Springer, Trois-Rivières, Québec, Canada, 2010, pp. 496–504.
[16] V.P.S. Naidu, Image fusion technique using multi-resolution singular value decomposition, Defence Sci. J. 61 (5) (2011) 479–484.
[17] B. Nolan, Data Analysis: An Introduction, Polity Press, UK, 1994.
[18] J. Shapiro, Embedded image coding using zerotrees of wavelet coefficients, IEEE Trans. Signal Process. 41 (12) (1993) 3445–3462.
[19] S.K. Singh, S. Kumar, Singular value decomposition based sub-band decomposition and multi-resolution (SVD-SBD-MRR) representation of digital colour images, Pertanika J. Sci. Technol. 19 (2) (2011) 229–235.
[20] J.C. Sprott, Chaos and Time-Series Analysis, Oxford University Press, USA, 2003.
[21] W. Sweldens, The lifting scheme: a construction of second generation wavelets, SIAM J. Math. Anal. 29 (2) (1998) 511–546.
[22] L. Yanhong, Q. Xiaohong, Speech signal enhancement based on adaptive multi-resolution form of SVD, in: Proc. Cong. Image and Signal Processing, IEEE, Sanya, Hainan, China, vol. 2, 2008, pp. 137–140.
[23] X. Zhao, B. Ye, T. Chen, Theory of multi-resolution singular value decomposition and its application to signal processing and fault diagnosis, J. Mech. Eng. 46 (20) (2010) 64–75.
[24] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE Trans. Image Process. 13 (4) (2004) 600–612.
[25] E. Drinea, P. Drineas, P. Huggins, A randomized singular value decomposition algorithm for image processing, in: Proc. Panhellenic Conf. Informatics, Springer, 2001, pp. 278–288.
[26] T. Bielecki, J. Chen, S.S.T. Yau, Random wavelet transformation and its properties, Proc. SPIE Wavelet Appl. Signal Image Process. II 2303 (1994) 345-1–345-10.
[27] http://cvc.yale.edu/projects/yalefaces/yalefaces.html.